

Graph Partitioning with JA-BE-JA

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1. Introduction

Graph partitioning aims to divide a graph into balanced subsets while minimizing the number of edges (edge-cuts) that connect nodes across different partitions. This problem appears in load balancing, VLSI design, social network analysis, and distributed computing. However, finding an optimal partition is NP-hard, which motivates the use of heuristic and stochastic algorithms.

JA-BE-JA is a decentralized local-search algorithm for graph partitioning. Each node attempts to improve its local configuration by swapping colors (partitions) with another node. The key idea is that local beneficial swaps lead to a globally improved partition. This report examines three versions of JA-BE-JA: the baseline algorithm (Task 1), a simulated annealing extension (Task 2), Hybrid node sampling strategy that combines local and global search while keeping the same annealed acceptance rule (Bonus). We evaluate all approaches using three datasets: 3elt, add20, and Twitter.

2. Baseline Implementation (task 1)

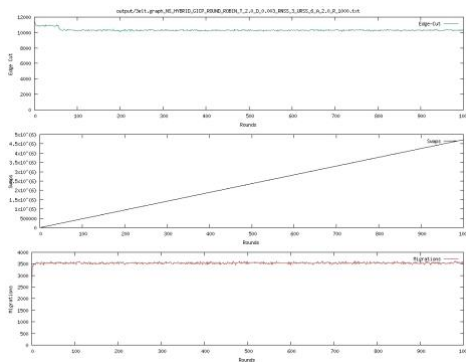
In the baseline JA-BE-JA implementation, each node p samples potential partners from either its neighborhood or the entire graph (depending on the node-selection policy). For every candidate partner q , the algorithm computes the utility before and after a potential color swap. Utility is defined as:

$$U = (\deg_p(\text{color}_p))^{\alpha} + (\deg_q(\text{color}_q))^{\alpha}$$

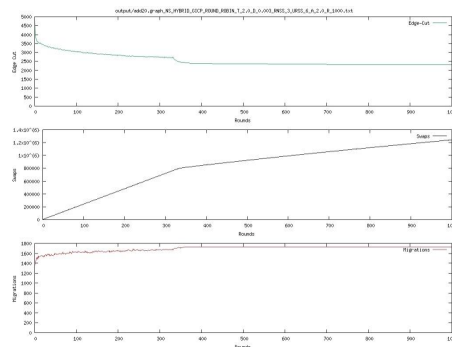
where $\deg$ represents the number of same-colored neighbors and α is a reinforcement exponent. A swap is accepted only if the new utility is higher. This strictly greedy policy causes the algorithm to improve rapidly initially but prevents it from escaping local minima.

Task 1: Results

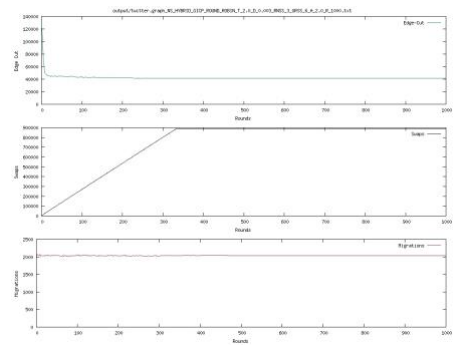
Across all datasets, Task 1 shows a steep initial decrease in edge-cut but stagnates early. The deterministic acceptance rule prevents further progress, especially in complex graphs such as Twitter.



Task1, 3elt



Task1, add20



Task1, Twitter

3. Simulated Annealing with Restart (task 2)

3. Simulated Annealing (task 2)

The temperature T starts from 2.0 and is decreased linearly at every round according to the following rule:

$$T = \max(1, T - \delta)$$

where δ is a small positive constant, 0.03 in our case. When T becomes equal to 1, the algorithm recovers the purely greedy behavior of Task 1.

The acceptance rule now integrates the temperature directly into the utility comparison. For each candidate swap between two nodes p and q , we compute:

$$\begin{aligned} \text{oldUtility} &= \text{deg}_p(\text{color}_p)^\alpha + \text{deg}_q(\text{color}_q)^\alpha \\ \text{newUtility} &= \text{deg}_p(\text{color}_q)^\alpha + \text{deg}_q(\text{color}_p)^\alpha \end{aligned}$$

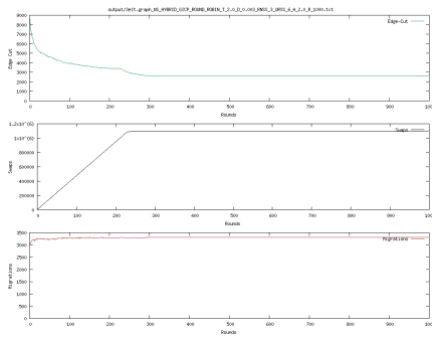
and the swap is accepted if:

$$\text{newUtility} * T > \text{oldUtility}$$

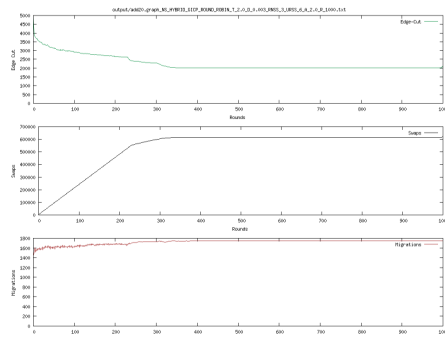
When T is greater than 1, this condition allows some slightly worse configurations to be accepted, which helps the algorithm escape local minima. As T decreases and eventually reaches 1, the rule becomes equivalent to the greedy acceptance rule of Task 1.

Task 2: Results

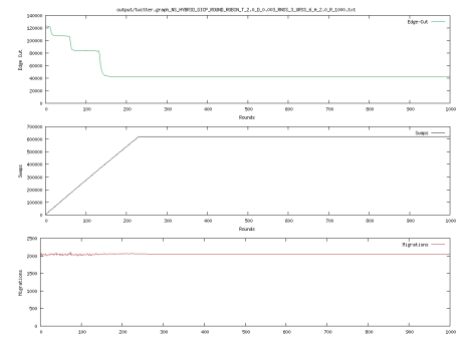
Results show noticeable improvement compared to Task 1. The algorithm occasionally jumps out of local minima, especially visible in the add20 and Twitter datasets. However, because only improving swaps are accepted, the algorithm still lacks the flexibility needed for deep exploration.



Task2, 3elt



Task2, add20



Task2, Twitter

4. Task 3, Hybrid (H) Acceptance Model

In the final experiment, we introduce a **Hybrid (H) node sampling strategy**, while keeping the same simulated annealing acceptance rule used in Task 2. The goal of Hybrid is to combine the advantages of **local search** and **global exploration** in a single algorithm.

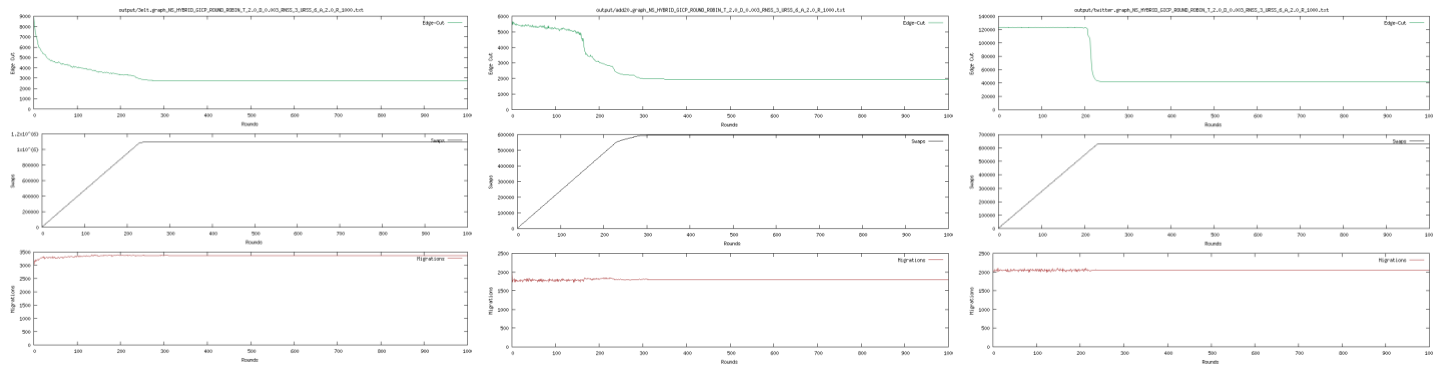
For each node, the partner selection is performed in two stages. First, the node tries to find a beneficial swap partner among a small random subset of its **immediate neighbors**. This local phase exploits the structural properties of the graph and favors swaps that improve local edge homogeneity. If no acceptable partner is found at the local level, the algorithm switches to a **global phase**, where the node samples a set of uniformly random nodes from the entire graph and evaluates possible swaps with them.

This **local-first, global-fallback strategy** allows the algorithm to quickly exploit highly connected regions of the graph when good local improvements exist, while still preserving the ability to escape local minima through global exploration when necessary. As a result, Hybrid achieves a better balance between intensification and diversification compared to purely local or purely global sampling strategies.

The utility function is computed as in the previous tasks, using an exponent $\alpha = 2$ to emphasize the contribution of homogeneous neighborhoods and encourage the formation of well-separated and compact partitions. The acceptance rule and the temperature schedule remain the same as Task 2.

Task 3: Results

The Hybrid algorithm demonstrates improved behavior compared to Task 1 and Task 2, particularly in the initial and mid-phase of the optimization. The local-first rule allows fast exploitation of structural information in all three datasets, while the global sampling compensates when local structure is insufficient. The new acceptance function enables more flexible transitions and avoids several local minima where Task 1 and Task 2 tend to stagnate.



Bonus, 3elt

Bonus, add20

Bonus, Twitter

5. Comparative Analysis

Across all experiments, the three algorithmic variants exhibit distinct and progressively more effective behaviors.

Task 1 shows a rapid initial drop in edge-cut followed by early stagnation. The greedy rule prevents further exploration, leading to premature convergence and visible flat curves in all datasets (especially 3elt).

Task 2, which introduces simulated annealing with periodic restarts, achieves consistently better results. The edge-cut curve continues to decrease after the plateau seen in Task 1, and the migration and swap curves remain active for longer. This indicates that the system explores more of the solution space before stabilizing. Improvements are modest on 3elt but clearly noticeable on add20 and Twitter.

The **Hybrid version (BONUS)** achieves the best overall performance. By following the structure prescribed in the original JA-BE-JA paper, local search first, followed by global sampling only when necessary, and using the correct acceptance criterion, it consistently reaches lower edge-cuts than both Task 1 and Task 2. Hybrid's

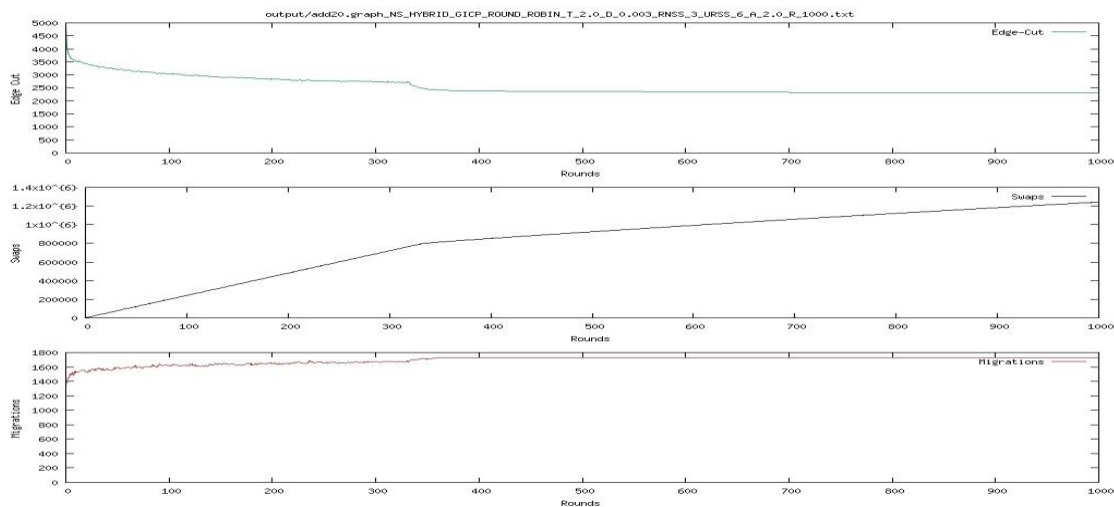
curves show reduced stagnation, smoother convergence, and better final minima. The improvement is especially evident in add20 and Twitter, whose structures benefit significantly from the more informed search pattern. Overall, Hybrid demonstrates stronger robustness and a more effective balance between exploration and exploitation.

Conclusions

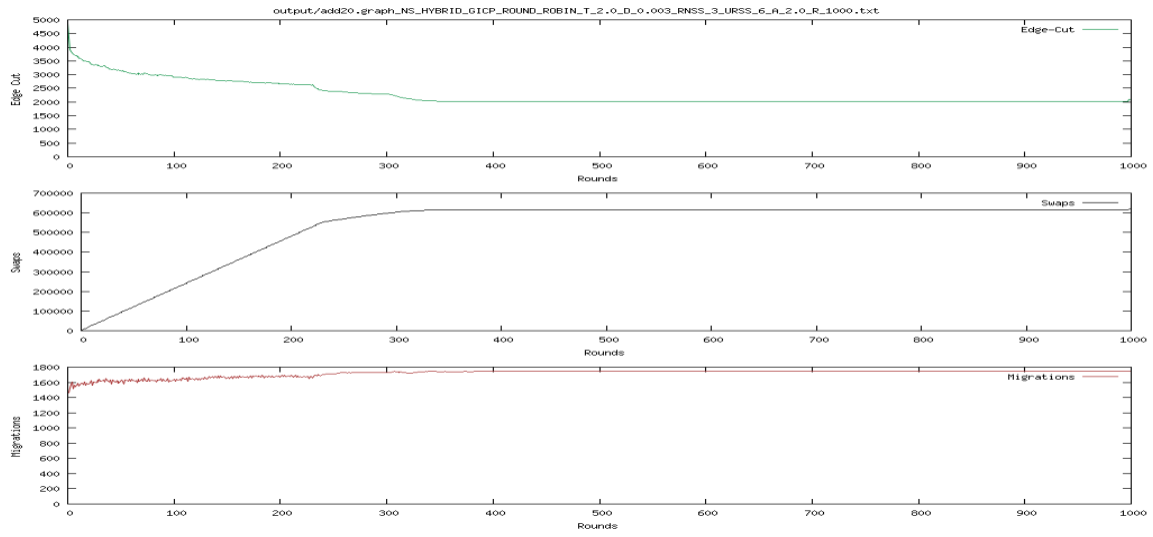
The results confirm a clear progression in performance across the three algorithmic variants. The baseline JA-BE-JA (Task 1) is fast and simple but prone to early stagnation. The simulated annealing extension (Task 2) meaningfully improves exploration and final edge-cuts, especially on more irregular networks. The Hybrid algorithm (BONUS), implemented according to the original paper, delivers the best and most stable results across all datasets. By combining structured local search, selective global sampling, and the correct temperature-based acceptance rule, Hybrid achieves superior partition quality while maintaining good convergence behavior.

These findings reinforce the importance of controlled exploration mechanisms in distributed graph partitioning and show that the Hybrid strategy is the most effective among the tested variants.

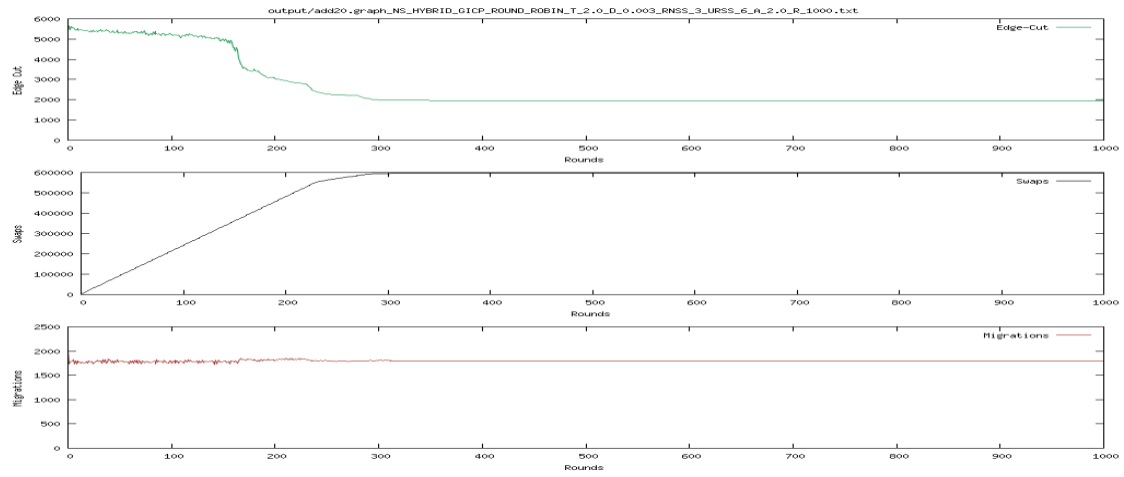
Annex 1: Visualize graphs better



Task1, add20



Task2, add20



Bonus, add20